Introduction

Along these tutorials, we will see how **SCOPE** interacts with the different parts of the code to handle the execution of computational workflows.

These are the topics covered in each tutorial:

1. The **System** class and its sources: the **Specie**, **Cell** and **Atom** classes
2. The Computational workflow: **Branch**, **Workflow**, **Job**, and **Computation** classes
3. The **State** class
4. The **Data**, **Collection** and **VNM** classes
5. The **Input_data** class, and **scope input files**
6. Running **SCOPE** - Part 1: File Structure
7. Running **SCOPE** - Part 2: Execution
8. Running **SCOPE** - Part 3: Detailed Actions
9. The Azo add-on: Subclasses, Analysis and Creation of Azo Systems

Tutorial 1: Systems and their Sources

In this tutorial, you will discover one of the main pillars of SCOPE: the **System**-class:

A **System** is way to organize collections of related chemical entities for later use in SCOPE.

- A **Source** is one of those chemical entities. There are two main types of sources: the **Cell** and the **Specie**:
- A **Specie** is an isolated chemical specie. It has various subclasses: being **Molecule** and **Ligand** the most important
- A **Cell** is a periodic structure. In essence, a Specie that includes cell parameters, but with different functionality.

In the first part of this tutorial, we will open an existing **SYSTEM**, stored in a binary file, and navigate the hierarchy of chemical species that it contains.
In the second part, we will create a new system from scratch, and feed it with **sources**

Loading [MathJax]/extensions/Safe.js

Part 1. Navigate an existing SYSTEM

In [1]:
```
import os
import scope
```

In [2]:
```
## Path of the data folder. It should be "os.path.abspath('../')+'/Data/1-Tu
data_folder = os.path.abspath('../')+'/Data/1-Tutorial_1/'

## Loads the System object from a binary file, provided in the tutorial folc
sys = scope.load_binary(f"{data_folder}/Systems/ABITEM/ABITEM.npy")

## This will set the main path of the system to the data folder, overwriting
## That way, any new file associated with this system created by SCOPE will
sys.set_main_path(data_folder)
```

Out[2]: True

In [3]:
```
## All objects in SCOPE have a __repr__ method, so printing shows a summary
print(sys)
```

```
--------------------------------------
-- >>> SCOPE System_sco Object >>> --
--------------------------------------
 Version                = 1.0
 Type                   = system
 Subtype                = sco_system
 Source Path            = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources            = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches           = 1
     0: isolated, Num Workflows: 2
```

The System is called ABITEM, and is associated with 6 **sources**:

```
0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
```

Loading [MathJax]/extensions/Safe.js

```
    3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
    4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
    5: specie, ref_ls_mol, H18-C20-N6-S2-Fe
```

Sources 0-1 are the **Cell** objects imported from the cell2mol interpretation of the .cif files

Sources 2-3 are the **Cell** objects identified as reference HS and LS unit cells.

Sources 4-5 are the **Specie** objects identified as reference HS and LS molecules.

Below, we will navigate these sources to discover the Cell and Specie classes

Sources 1: The CELL Class

In [4]:
```
# You can find sources by name, with the dedicated system-class function
found, cell = sys.find_source("ABITEM")
print('Found:', found)
# All objects in SCOPE can be printed, which displays their key information.
print(cell)
```

```
Found: True
-----------------------------------
-- >>> SCOPE Cell_sco Object >>> --
-----------------------------------
 Version               = 1.0
 Type                  = cell
 SubType               = sco_cell
 Name                  = ABITEM
 Num Atoms             = 236
 Cell Parameters a:c   = [12.3008, 11.9241, 17.6916]
 Cell Parameters al:ga = [90.0, 92.704, 90.0]
 Volume (Angs^3)       = 2592.0433
 Number of Units (Z)   = 0
 Num Molecules:        = 8
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H18-C20-N6-S2-Fe
    2: H8-C3-O
    3: H8-C3-O
    4: H8-C3-O
    5: H8-C3-O
    6: H18-C20-N6-S2-Fe
    7: H18-C20-N6-S2-Fe
-------------------------------
 Num of Ref Molecules: = 2
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H8-C3-O
 Phase                 = HS
 HS Molar Fraction     = 1.0
```

In [5]:
```
# Have a look at the CELL object above.
```

Loading [MathJax]/extensions/Safe.js

```
# -It has 236 atoms, whose coordinates are stored as cell.coord, and labels
print("Labels and Coordinates:", cell.labels[0], cell.coord[0])
# -It has cell parameters, stored as cell.cell_param
print("Cell Parameters: (a,b,c,alpha,beta,gamma)", cell.cell_param)
# -It has 8 Molecules, stored inside cell.molecules
print("Number of Molecules:", len(cell.molecules))
# -It has 2 unique (i.e. types of) molecules, with formulae, and occurrence
for mol in cell.ref_molecules:
    print("Unique Molecule Formula:", mol.formula, cell.get_occurrence(mol))
# -It has 4 stoichiometry units in the Cell (Z). That is, the stoichiometry
print("Number of Stoichiometry Units (Z):", cell.z)
```

```
Labels and Coordinates: Fe [3.7430176, 3.9944543, 13.5094621]
Cell Parameters: (a,b,c,alpha,beta,gamma) [12.3008, 11.9241, 17.6916, 90.0,
92.704, 90.0]
Number of Molecules: 8
Unique Molecule Formula: H18-C20-N6-S2-Fe 0
Unique Molecule Formula: H8-C3-O 0
Number of Stoichiometry Units (Z): 0
```

In [6]:

```
# Apart from attributes, you can also run CELL-class functions.
# Here we try cell.check_fragmentation(), which checks if any molecule appea
cell.check_fragmentation(debug=1)
```

Loading [MathJax]/extensions/Safe.js

```
CELL.CHECK_FRAGMENTATION: molecules available. Checking Fragmentation
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H18-C20-N6-S2-Fe wit
hout graph analysis: issame=True
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H8-C3-O without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H18-C20-N6-S2-Fe wit
hout graph analysis: issame=True
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H8-C3-O without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H18-C20-N6-S2-Fe without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H8-C3-O without graph analysi
s: issame=True
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H18-C20-N6-S2-Fe without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H8-C3-O without graph analysi
s: issame=True
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H18-C20-N6-S2-Fe without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H8-C3-O without graph analysi
s: issame=True
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H18-C20-N6-S2-Fe without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H8-C3-O and H8-C3-O without graph analysi
s: issame=True
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H18-C20-N6-S2-Fe wit
hout graph analysis: issame=True
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H8-C3-O without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H18-C20-N6-S2-Fe wit
hout graph analysis: issame=True
CELL.CHECK_FRAGMENTATION: compared H18-C20-N6-S2-Fe and H8-C3-O without grap
h analysis: issame=False
CELL.CHECK_FRAGMENTATION: cell is not fragmented
```

Out[6]: False

In [7]:
```
# This CELL is a little bit special, though. It was created by the "Spin_Cro
# This add-on creates SCO_CELLs, which are subclasses of CELL, created to ac
# Notice the subtype
print(cell.object_type)
print(cell.object_subtype)
```

```
cell
sco_cell
```

In [8]:
```
# This CELL was created from cell2mol data, and is associated with a CIF fil
print(cell.cif)
```

Loading [MathJax]/extensions/Safe.js

```
----------------------------------------------------
                    SCOPE .Cif file
----------------------------------------------------
 Name                  = ABITEM.cif
 Path                  = /data/ucsscms/svela/4-Merged/ABITEM/Original/ABITE
M.cif
 Symmetry Group        = P 21/c
 Symmetry Operations   = ['x,y,z', '-x,1/2+y,1/2-z', '-x,-y,-z', 'x,-1/2-y,-
1/2+z']
 Diffraction Temp      = 240.0
 Authors               = ['Rong-Jia Wei', 'Jun Tao', 'Rong-Bin Huang', 'Lan-
Sun Zheng']
 Year of Publication   = 2011
 Journal Name          = Inorg.Chem.
 Journal Volume        = 50
 Journal Page          = 8553
 Has Associated Cell   = YES
```

In [9]:
```
# You can view the unit cell with cell.view(), which opens a 3D viewer
cell.view(size='large')
```

Sources 2: The SPECIE Class

In [10]:
```
# Hopefully, you can identify the 4 transition metal complexes (TMCs) and th
# In any case, you can access those molecules through the cell.molecules att
len(cell.molecules)
```

Out[10]: 8

Molecule Sub-Class

In [11]:
```
# MOLECULE-class objects are a subclass of SPECIE, which more broadly refers
# Here, we take the first molecule in the unit cell as an example:
mol = cell.molecules[0]
# Again, the key information can be displayed by printing the object
print(mol)
```

Loading [MathJax]/extensions/Safe.js

```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version                = 1.0
 Type                   = specie
 Sub-Type               = molecule
 Number of Atoms        = 47
 Formula                = H18-C20-N6-S2-Fe
 Charge                 = 0
 Spin (alpha - beta)    = 0
 Number of Parents      = 1
 Has Adjacency Matrix   = YES
 Has Bonds              = YES

 Num of Ligands         = 3
   Ligand 0: C-N-S with 3 atoms. Smiles: [N-]=C=S
   Ligand 1: H18-C18-N4 with 40 atoms. Smiles: [H]c1nc(C([H])([H])N(C([H])
([H])c2nc([H])c([H])c([H])c2[H])C([H])([H])c2nc([H])c([H])c([H])c2[H])c([H])
c([H])c1[H]
   Ligand 2: C-N-S with 3 atoms. Smiles: [N-]=C=S

 Num of Metals          = 1
   Metal 0: Fe with charge:2 and spin:0
```

In [12]:
```
# Notice the molecule is of type=SPECIE, and subtype=MOLECULE. As mentioned
# Also, this molecule is a TMC, as shown by:
print(mol.iscomplex)
# SCOPE will flag a molecule as a TMC whenever it has any atom of the d- or
# Whenever this happens, it will automatically split the molecule into metal
```

```
True
```

In [13]:
```
# Ligands can be accessed with the "ligands" attribute:
print(len(mol.ligands), "ligands in this TMC")
```

```
3 ligands in this TMC
```

In [14]:
```
# Similarly, Metals can be accessed with the "metals" attribute:
print(len(mol.metals), "metal atom(s) in this TMC")
# Metals belong to the ATOM class, which we will soon discover
```

```
1 metal atom(s) in this TMC
```

In [15]:
```
## All species can be visualized with the .view() method:
mol.view()
```

Comparing Species

In [16]:
```
### Also, SPECIES can be compared using the == operator:
mol2 = cell.molecules[1]
print(mol == mol2)

## The comparison above checks the composition of the molecules, but not the
```

Loading [MathJax]/extensions/Safe.js

```
## Thus, two conformers of the same molecule will be considered equal.
## The comparison uses chemical composition, connectivity, and graph topolog
```

```
True
```

In [17]:
```
import time
import numpy as np

## Here, you can compare all molecules in the unit cell.
for idx, m1 in enumerate(cell.molecules):
    for jdx, m2 in enumerate(cell.molecules):
        # If they are the same, their RMSD is also calculated.
        if m1 == m2 and jdx > idx:
            tini = time.time()
            rmsd = m1.rmsd(m2)
            tend = time.time()
            elapsed = np.round(tend - tini,2)
            print(idx, jdx, m1.formula, m1 == m2, "RMSD:", rmsd, "in", elaps
        #elif jdx > idx:                                        # un-comment
        #    print(idx, jdx, m1.formula, m1 == m2)

# The computation of the RMSD requires that molecules are re-aligned, and th
# ... to achieve the best possible overlap. This is done internally by SCOPE
#  -- scope.other.overlap_molecules() --
# Once overlapped, the RMSD can be easily computed:
```

```
0 1 H18-C20-N6-S2-Fe True RMSD: 3.58 in 0.1 seconds
0 6 H18-C20-N6-S2-Fe True RMSD: 0.0001 in 0.02 seconds
0 7 H18-C20-N6-S2-Fe True RMSD: 2.2408 in 0.02 seconds
1 6 H18-C20-N6-S2-Fe True RMSD: 2.2408 in 0.02 seconds
1 7 H18-C20-N6-S2-Fe True RMSD: 0.0001 in 0.02 seconds
2 3 H8-C3-O True RMSD: 0.1841 in 0.0 seconds
2 4 H8-C3-O True RMSD: 0.1841 in 0.0 seconds
2 5 H8-C3-O True RMSD: 0.0 in 0.0 seconds
3 4 H8-C3-O True RMSD: 0.0 in 0.0 seconds
3 5 H8-C3-O True RMSD: 0.1841 in 0.0 seconds
4 5 H8-C3-O True RMSD: 0.1841 in 0.0 seconds
6 7 H18-C20-N6-S2-Fe True RMSD: 2.2408 in 0.02 seconds
```

In [18]:
```
## Notice that the RMSD comparison above reflects the symmetry of the molecu

## As you can see in the .cif file printed above, this Cell has P 21/c symme
print(cell.cif.sym_group)
## This symmetry group is associated with 4 symmetry operations.
print(cell.cif.sym_ops)
```

```
P 21/c
['x,y,z', '-x,1/2+y,1/2-z', '-x,-y,-z', 'x,-1/2-y,-1/2+z']
```

In [19]:
```
# Leaving the identity aside, there are three symmetry operations connecting
# As a result, there are three groups of RMSD values within the set of TMC:

# For the EtOH molecule, its higher internal symmetry overcomes the crystall
# resulting in only two groups of RMSD values [0.0, 0.1841]
```

In [20]:
```
## The same task (overlap + RMSD) can be performed with RDKIT.
## Below is a basic script to build the molecule objects from the coordinate
```

Loading [MathJax]/extensions/Safe.js

```
## And then do the RMSD computation, carried out by the rdMolAlign.GetBestRM

from rdkit import Chem
from rdkit.Chem import rdMolAlign

def build_rdkit_mol(atom_labels, coords):
    """
    atom_labels: list of strings, e.g. ['C', 'H', 'H', 'O']
    coords: np.array of shape (N_atoms, 3)
    """
    mol = Chem.RWMol()  # editable molecule
    # add atoms
    for label in atom_labels:
        atom = Chem.Atom(label)
        mol.AddAtom(atom)
    # create a conformer (3D structure)
    conf = Chem.Conformer(len(atom_labels))
    for i, (x, y, z) in enumerate(coords):
        conf.SetAtomPosition(i, Chem.rdGeometry.Point3D(x, y, z))
    mol.AddConformer(conf)
    return mol

## Againg, we compare all molecules in the unit cell.
for idx, m1 in enumerate(cell.molecules):
    for jdx, m2 in enumerate(cell.molecules):
        # If they are the same, their RMSD is also calculated.
        if m1 == m2 and jdx > idx:
            a1 = build_rdkit_mol(m1.labels, m1.coord)
            a2 = build_rdkit_mol(m2.labels, m2.coord)
            tini = time.time()
            rmsd_rdkit = np.round(rdMolAlign.GetBestRMS(a1, a2),2)
            tend = time.time()
            elapsed = np.round(tend - tini,2)
            print(idx, jdx, m1.formula, m1 == m2, "RMSD:", rmsd_rdkit, "in",
```

```
0 1 H18-C20-N6-S2-Fe True RMSD: 2.97 in 2.14 seconds
0 6 H18-C20-N6-S2-Fe True RMSD: 3.46 in 2.09 seconds
0 7 H18-C20-N6-S2-Fe True RMSD: 4.17 in 2.14 seconds
1 6 H18-C20-N6-S2-Fe True RMSD: 3.95 in 2.19 seconds
1 7 H18-C20-N6-S2-Fe True RMSD: 3.39 in 2.18 seconds
2 3 H8-C3-O True RMSD: 0.18 in 0.32 seconds
2 4 H8-C3-O True RMSD: 0.18 in 0.31 seconds
2 5 H8-C3-O True RMSD: 0.0 in 0.31 seconds
3 4 H8-C3-O True RMSD: 0.0 in 0.32 seconds
3 5 H8-C3-O True RMSD: 0.18 in 0.31 seconds
4 5 H8-C3-O True RMSD: 0.18 in 0.32 seconds
6 7 H18-C20-N6-S2-Fe True RMSD: 3.22 in 2.07 seconds
```

In [21]:

```
## Notice that the evaluation of the RMSD takes 10 times that of SCOPE
## Also, the crystal symmetry, which was succesfully captured in the RMSD va
## Manual inspection reveals that this is because the overlap procedure in R
```

Ligand Sub-Class

Loading [MathJax]/extensions/Safe.js

```
you can access the ligand objects through the "ligands" attribute
lig = mol.ligands[0]
```

```
print(lig)
```

```
-----------------------------------
------- SCOPE LIGAND Object -------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = ligand
 Number of Atoms       = 3
 Formula               = C-N-S
 Charge                = -1
 Spin (alpha - beta)   = 0
 SMILES                = [N-]=C=S
 Number of Parents     = 2
 Has Adjacency Matrix  = NO
 Has Bonds             = YES
 Has RDKIT Object      = YES
```

In [23]:
```
## The LIGAND is also a subclass of SPECIE (object_type=specie, object_subty
print(lig.object_type, lig.object_subtype)
## It has some useful functions too, e.g.:
print(lig.get_denticity())        ## This one retrieves the denticity of the
print(lig.get_connected_metals()) ## This one retreives all metals to which
```

```
specie ligand
1
[-----------------------------------------------
------------- SCOPE METAL Object --------------
-----------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = metal
 Label                        = Fe
 Atomic Number                = 26
 Atom Charge                  = 2
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 6
 Regular Adjacencies (adjnum) = 6

]
```

In [24]:
```
## As mentioned above, the Cell objects in this tutorial were imported from
## In these Cells, Ligands are associated with an rdkit object...
lig.rdkit_obj

## As you can see, this object is imported and corrected: there were some is
```

Loading [MathJax]/extensions/Safe.js

Out[24]:

In [25]:
```
## ... and a Smiles
lig.smiles
```

Out[25]: '[N-]=C=S'

Metal Sub-Class

In [26]:
```
## Again, METAL is a subclass of ATOM (type=atom, subtype=metal)
## You can access the METAL object from the ligand with (1) lig.get_connecte
met = mol.metals[0]
print(met)
```

```
-----------------------------------------------
------------- SCOPE METAL Object --------------
-----------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = metal
 Label                        = Fe
 Atomic Number                = 26
 Atom Charge                  = 2
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 6
 Regular Adjacencies (adjnum) = 6
```

In [27]:
```
## Apart from attributes, it also has useful functions, e.g.:
print(met.get_coord_sphere())          ## These are the atoms that are coord
```

Loading [MathJax]/extensions/Safe.js

```
[-------------------------------------------------
------------- SCOPE ATOM Object ----------------
-------------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = atom
 Label                        = N
 Atomic Number                = 7
 Atom Charge                  = 0
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 1
 Regular Adjacencies (adjnum) = 3

, -------------------------------------------------
------------- SCOPE ATOM Object ----------------
-------------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = atom
 Label                        = N
 Atomic Number                = 7
 Atom Charge                  = 0
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 1
 Regular Adjacencies (adjnum) = 3

, -------------------------------------------------
------------- SCOPE ATOM Object ----------------
-------------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = atom
 Label                        = N
 Atomic Number                = 7
 Atom Charge                  = 0
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 1
 Regular Adjacencies (adjnum) = 3

, -------------------------------------------------
------------- SCOPE ATOM Object ----------------
-------------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = atom
 Label                        = N
 Atomic Number                = 7
 Atom Charge                  = 0
 Spin (alpha - beta)          = 0
 Metal Adjacencies (madjnum)  = 1
 Regular Adjacencies (adjnum) = 4

, -------------------------------------------------
------------- SCOPE ATOM Object ----------------
-------------------------------------------------
 Version                      = 1.0
```

Loading [MathJax]/extensions/Safe.js

```
 Type                          = atom
 Sub-Type                      = atom
 Label                         = N
 Atomic Number                 = 7
 Atom Charge                   = -2
 Spin (alpha - beta)           = 0
 Metal Adjacencies (madjnum)   = 1
 Regular Adjacencies (adjnum) = 2

, ------------------------------------------------
------------- SCOPE ATOM Object ----------------
------------------------------------------------
 Version                       = 1.0
 Type                          = atom
 Sub-Type                      = atom
 Label                         = N
 Atomic Number                 = 7
 Atom Charge                   = -2
 Spin (alpha - beta)           = 0
 Metal Adjacencies (madjnum)   = 1
 Regular Adjacencies (adjnum) = 2

]
```

In [28]:
```
print(met.get_coord_sphere_formula())  ## This is the formula of the atoms i
```

```
N6
```

In [29]:
```
#met.get_cshm()   ## This one retrieves the Coordination Shape Measure of th
                  ## The CShM quantifies how close the coordination environm
                  ## Unfortunately, the package that computes it (Cosymlib)
                  ## Those dependencies, related to numpy, make the installa
                  ## We are currently waiting for an update of Cosymlib addr
                  ## In the meantime, if you're brave enough, you can try to
```

Going Upwards: Parents

In [30]:
```
## So far, we went down in the hierarchy: System -> Cell -> Molecule -> Meta

# This retrieves the parent CELL object of the ligand
parent = lig.get_parent('cell')
print(parent)
```

Loading [MathJax]/extensions/Safe.js

```
-------------------------------
   >>> SCOPE CELL Object >>>
-------------------------------
 Version               = 1.0
 Type                  = cell
 SubType               = cell
 Name                  = ABITEM
 Num Atoms             = 236
 Cell Parameters a:c   = [12.3008, 11.9241, 17.6916]
 Cell Parameters al:ga = [90.0, 92.704, 90.0]
 Volume (Angs^3)       = 2592.0433
 Number of Units (Z)   = 0
 Num Molecules:        = 8
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H18-C20-N6-S2-Fe
    2: H8-C3-O
    3: H8-C3-O
    4: H8-C3-O
    5: H8-C3-O
    6: H18-C20-N6-S2-Fe
    7: H18-C20-N6-S2-Fe
-------------------------------
 Num of Ref Molecules: = 2
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H8-C3-O
```

In [31]:

```
# And this retrieves the index of the ligand atoms in the parent molecule, s
idx = lig.get_parent_indices('molecule')
print("Ligand Atom Indices in Parent's Molecule class:", idx)

for i, pi in enumerate(idx):
    lig_atom = lig.atoms[i]                                          ## i
    mol_atom = lig.get_parent('molecule').atoms[pi]                  ## p
    print(i, pi, lig_atom.label, mol_atom.label, lig_atom == mol_atom)  ## H
```

```
Ligand Atom Indices in Parent's Molecule class: [2, 8, 42]
0 2 S S True
1 8 N N True
2 42 C C True
```

The Elephant in the Room: The ATOM Class

In [32]:

```
# As we've briefly seen above, species have atoms.
len(mol.atoms)
```

Out[32]: 47

In [33]:

```
## The first atom of the molecule happens to be the metal. The second is a r
for at in mol.atoms[0:5]:                                   ## Printing the f
    print(at.get_parent_index('molecule'), at.label, at.object_subtype, at.c
```

Loading [MathJax]/extensions/Safe.js

```
## at.charge    is the formal atomic charge.
```

```
## at.madjnum   is the number of adjacencies the atom has with a metal cente
## at.adjnum    is the total number of adjacencies the atom has with any oth
```

```
0 Fe metal 2 6 6
1 S atom 1 0 1
2 S atom 1 0 1
3 N atom 0 1 3
4 N atom 0 1 3
```

Charge and Spin

In [34]:
```
## Because cell2mol infers the charge of species in a unit cell, this info i
for mol in cell.molecules:
    print(f"Formula: {mol.formula}, Charge: {mol.charge}, Spin: {mol.spin}")
```

```
Formula: H18-C20-N6-S2-Fe, Charge: 0, Spin: 0
Formula: H18-C20-N6-S2-Fe, Charge: 0, Spin: 0
Formula: H8-C3-O, Charge: 0, Spin: 0
Formula: H8-C3-O, Charge: 0, Spin: 0
Formula: H8-C3-O, Charge: 0, Spin: 0
Formula: H8-C3-O, Charge: 0, Spin: 0
Formula: H18-C20-N6-S2-Fe, Charge: 0, Spin: 0
Formula: H18-C20-N6-S2-Fe, Charge: 0, Spin: 0
```

In [35]:
```
## And also for the ligands and metals inside the molecules:
mol = cell.molecules[0]
for lig in mol.ligands:
    print(f"Ligand Formula: {lig.formula}, Charge: {lig.charge}, Spin: {lig.
for met in mol.metals:
    print(f"Metal: {met.label}, Charge: {met.charge}, Spin: {met.spin}")
```

```
Ligand Formula: C-N-S, Charge: -1, Spin: 0
Ligand Formula: H18-C18-N4, Charge: 0, Spin: 0
Ligand Formula: C-N-S, Charge: -1, Spin: 0
Metal: Fe, Charge: 2, Spin: 0
```

In [36]:
```
## All the way down to the ATOM objects, which are the actual bearers of spi
for at in mol.atoms:
    print(f"Atom: {at.label}, Charge: {at.charge}, Spin: {at.spin}")
```

Loading [MathJax]/extensions/Safe.js

```
Atom: Fe, Charge: 2, Spin: 0
Atom: S, Charge: 1, Spin: 0
Atom: S, Charge: 1, Spin: 0
Atom: N, Charge: 0, Spin: 0
Atom: N, Charge: 0, Spin: 0
Atom: N, Charge: 0, Spin: 0
Atom: N, Charge: 0, Spin: 0
Atom: N, Charge: -2, Spin: 0
Atom: N, Charge: -2, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
Atom: C, Charge: 0, Spin: 0
Atom: H, Charge: 0, Spin: 0
```

In [37]:

```
## IMPORTANT: The charge and spin attribute for the SPECIE (molecules, ligar
## ... as the sum of the charge and spin of their constituent ATOM objects.m

## - If you want to modify an ATOM charge/spin, you can do it on the atom ob
##   - at.set_charge(1)
##   - at.set_spin(2)
##     Alternatively, you can use spec.set_atomic_charges([list_of_charges])
##     ... and spec.set_atomic_spins([list_of_spins]) to set all atomic char
```

Loading [MathJax]/extensions/Safe.js

```
## Notice that, by "spin" we mean the total number of unpaired electrons (al
## For instance, a spin of 1 means one unpaired electron (doublet).

## Here we take atom 0, which should be an Fe atom,
at = mol.atoms[0]

## We set it as a high-spin center, with 4 unpaired electrons (spin=4)
at.set_spin(4)

## When introducing the spin, the multiplicity and ms values are also stored
print(f"Spin: {at.spin}, Ms: {at.ms}, Multiplicity: {at.multiplicity}")

## When Working with the STATE class in tutorial number 2, we will see how w
```

```
Spin: 4, Ms: 2.0, Multiplicity: 5
```

Bond Class

In [38]:
```
## Bonds are also stored as objects, with very limited functionality for now
print(at.bonds)
## The formal bond order (bond.order) value above is extracted from the rdki
```

Loading [MathJax]/extensions/Safe.js

```
[-----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 8
 Label Atom1             = Fe
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.111 Angstrom

, -----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 3
 Label Atom1             = Fe
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.189 Angstrom

, -----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 4
 Label Atom1             = Fe
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.202 Angstrom

, -----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 5
 Label Atom1             = Fe
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.202 Angstrom

, -----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 6
 Label Atom1             = Fe
```

Loading [MathJax]/extensions/Safe.js

```
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.238 Angstrom

, ----------------------------------------------
------------- SCOPE BOND Object --------------
----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 7
 Label Atom1             = Fe
 Label Atom2             = N
 Formal Bond Order       = 0.5
 Distance                = 2.063 Angstrom

]
```

Part 2. Create a System

We will now create a simple system, that we will use later during Tutorial 7.

Systems only require a name, and are used mainly for organizational purposes, as we discussed above:

- A **System** is a collection of chemical entities with something in common, typically its chemical composition. The Main Purpose of this class is to store paths for later use in SCOPE, as well as its **sources**.

In [39]:
```
## Path of Tutorial 7's Data folder. Normally, it should be "os.path.abspath
tutorial_folder = os.path.abspath('../')+'/Data/7-Tutorial_7/'
```

In [40]:
```
## To create a system, you only need a name
test_sys   = scope.System("water")
print(test_sys)
```

```
--------------------------------
   >>> SCOPE System Object >>>
--------------------------------
 Version               = 1.0
 Type                  = system
 Subtype               = system
```

In [41]:
```
## The created system is still 'empty'. That is, no sources are associated t
## Before feeding it with sources, we will set the paths of the three releva
## - Sources         (test_sys.sources_path)
## - Computations    (test_sys.computations_path)
## - System          (test_sys.system_path)
```

Loading [MathJax]/extensions/Safe.js

In [42]:
```
## You can do it in 3 ways:

## (1) Directly as:
test_sys.sources_path      = f"{tutorial_folder}Sources/water/"
test_sys.system_path       = f"{tutorial_folder}Systems/water/"
test_sys.system_file       = f"{tutorial_folder}Systems/water/water.npy"
test_sys.computations_path = f"{tutorial_folder}Computations/water/"

## (2) Reading from an Environment-class object (uncomment to use). If you'r
# First, you load (or create) the enviroment.
#   env = load_binary(path)
# Second:
#   test_sys.set_paths_from_environment(env)

## (3) With test_sys.set_paths(), and you just need to follow the instructic
```

In [43]:
```
## Once done, you can print the system to check that everything is correct
print(test_sys)

## You can check whether the paths are correct with the function below. It w
print(test_sys.check_paths(debug=1))
```

```
---------------------------------
   >>> SCOPE System Object >>>
---------------------------------
 Version               = 1.0
 Type                  = system
 Subtype               = system
 Source Path           = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Sources/water/
 System File Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Systems/water/
 System File Name      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Systems/water/water.npy
 Computations Path     = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Computations/water/


True
```

In [44]:
```
## At this stage, WARNINGS might pop up because some folder might not exist
## Folders are automatically created with options (2) and (3). So don't worr
```

Create a Molecule and Associate it to the System (i.e. Sourcing)

- Currently, sources can be imported from three formats:
  1. XYZ file
  2. RDKIT molecule object
  3. CELL2MOL cell -or molecule- object (information on cell2mol can be found in: 10.1038/s41524-022-00874-9)

Loading [MathJax]/extensions/Safe.js

Below, we will (1) import a molecule from an XYZ file, and source it to our "water" system.
Also, we will explore the options of sourcing from either (2) an RDKIT molecule object, or (3) a cell2mol Cell object.

We will only keep (1) for our water system. (2) and (3) are only shown for completeness.

1) Sourcing from an XYZ file

In [45]:
```
## Now we create an actual water molecule, whose geometry is read from an .x
from scope.read_write     import read_xyz
from scope.classes_specie import Molecule

labels, coords = read_xyz(test_sys.sources_path+"geom1.xyz")
molec          = Molecule(labels, coords)
print(molec)
```

```
------------------------------------
------ SCOPE MOLECULE Object ------
------------------------------------
 Version                = 1.0
 Type                   = specie
 Sub-Type               = molecule
 Number of Atoms        = 3
 Formula                = H2-O
 Charge                 = 0
 Spin (alpha - beta)    = 0
 Number of Parents      = 0
 Has Adjacency Matrix   = NO
 Has Bonds              = NO
```

2) Sourcing from an RDKIT molecule

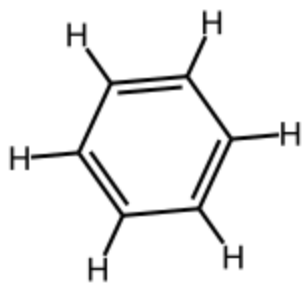
In [46]:
```
## If we were to source an rdkit molecule object, we would do the following:
from rdkit import Chem

rdkit_mol       = Chem.MolFromSmiles('c1ccccc1')                        ## A
imported_mol    = scope.classes_specie.import_molecule(rdkit_mol)       ## I
print(imported_mol)

## The function scope.classes_specie.import_molecule() is used to import mol
## Basically, as long as compatibility is possible, you will be able to upda
```

Loading [MathJax]/extensions/Safe.js

```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = molecule
 Number of Atoms       = 12
 Formula               = H6-C6
 Charge                = 0
 Spin (alpha - beta)   = 0
 SMILES                = [H]c1c([H])c([H])c([H])c([H])c1[H]
 Number of Parents     = 0
 Has Adjacency Matrix  = YES
 Has Bonds             = YES
```

In [47]:
```
# SMILES strings encode the Lewis Structure of the molecule.
# When creating the molecule, RDKIT stores this information.
# In SCOPE, when importing RDKIT molecules, the rdkit object itself is store
imported_mol.rdkit_obj
```

Out[47]:



In [48]:
```
# When rdkit molecules are imported, Bond-class instances are generated auto
# For instance, this is the first bond in which the first atom of the molecu
print(imported_mol.atoms[0].bonds[0])
```

```
-----------------------------------------------
------------- SCOPE BOND Object --------------
-----------------------------------------------
 Version                 = 1.0
 Type                    = bond
 Atom1 Index in Molecule = 0
 Atom2 Index in Molecule = 1
 Label Atom1             = C
 Label Atom2             = C
 Formal Bond Order       = 1.5
 Distance                = 1.399 Angstrom
```

3) Sourcing from cell2mol

In [48]:
```
## A CELL object is provided as part of this tutorials
cell2mol_cell = scope.load_binary(f"{sys.sources_path}Cell_ABITEM.gmol")

ately, cell2mol objects do not (yet) have any print method to sum
```

Loading [MathJax]/extensions/Safe.js

```
for cell2mol_mol in cell2mol_cell.moleclist:
    print(cell2mol_mol.formula)
```

```
H18-C20-N6-S2-Fe
H18-C20-N6-S2-Fe
H8-C3-O
H8-C3-O
H8-C3-O
H8-C3-O
H18-C20-N6-S2-Fe
H18-C20-N6-S2-Fe
```

In [49]:
```
## Similar to what we did before, the function scope.classes_cell.import_cel
## ... or previous versions of SCOPE.
## Here we import the unit cell:
imported_cell = scope.classes_cell.import_cell(cell2mol_cell)
print(imported_cell)
```

```
-------------------------------
   >>> SCOPE CELL Object >>>
-------------------------------
 Version               = 1.0
 Type                  = cell
 SubType               = cell
 Name                  = ABITEM
 Num Atoms             = 236
 Cell Parameters a:c   = [12.3008, 11.9241, 17.6916]
 Cell Parameters al:ga = [90.0, 92.704, 90.0]
 Volume (Angs^3)       = 2592.0433
 Number of Units (Z)   = 0
 Num Molecules:        = 8
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H18-C20-N6-S2-Fe
    2: H8-C3-O
    3: H8-C3-O
    4: H8-C3-O
    5: H8-C3-O
    6: H18-C20-N6-S2-Fe
    7: H18-C20-N6-S2-Fe
-------------------------------
 Num of Ref Molecules: = 2
 With Formulae:
    0: H18-C20-N6-S2-Fe
    1: H8-C3-O
```

In [50]:
```
## Here we import the one of the molecules in the unit cell:
imported_mol = scope.classes_specie.import_molecule(cell2mol_mol)
print(imported_mol)
```

Loading [MathJax]/extensions/Safe.js

```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = molecule
 Number of Atoms       = 47
 Formula               = H18-C20-N6-S2-Fe
 Charge                = 0
 Spin (alpha - beta)   = 0
 Number of Parents     = 0
 Has Adjacency Matrix  = YES
 Has Bonds             = NO

 Num of Ligands        = 3
   Ligand 0: C-N-S with 3 atoms. Smiles: [N-]=C=S
   Ligand 1: H18-C18-N4 with 40 atoms. Smiles: [H]c1nc(C([H])([H])N(C([H])
([H])c2nc([H])c([H])c([H])c2[H])C([H])([H])c2nc([H])c([H])c([H])c2[H])c([H])
c([H])c1[H]
   Ligand 2: C-N-S with 3 atoms. Smiles: [N-]=C=S

 Num of Metals         = 1
   Metal 0: Fe with charge:2 and spin:0
```

Charge and Spin

In [51]:
```
## By default, molecules are neutral (and singlets) when created.
## You can change it if needed:
molec.set_total_charge(-1)
## Notice that the charge has changed:
molec
```

Out[51]:
```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = molecule
 Number of Atoms       = 3
 Formula               = H2-O
 Charge                = -1
 Spin (alpha - beta)   = 0
 Number of Parents     = 0
 Has Adjacency Matrix  = NO
 Has Bonds             = NO
```

In [52]:
```
## molec.set_total_charge() and molec.set_total_spin() are enough if you jus
## These functions just assign the desired charge/spin to the first atom of
molec.atoms[0]
```

Loading [MathJax]/extensions/Safe.js

Out[52]:
```
-------------------------------------------------
------------- SCOPE ATOM Object -----------------
-------------------------------------------------
 Version                      = 1.0
 Type                         = atom
 Sub-Type                     = atom
 Label                        = O
 Atomic Number                = 8
 Atom Charge                  = -1
 Spin (alpha - beta)          = 0
 No Metal Adjacency Info
 No Adjacency Info
```

In [53]:
```
## In this case, the first atom happens to be an Oxigen, so the -1 charge is
## If it were a Hydrogen, it would be strange to have a -1 charge on it.

## So it is much better if you individually select the atom you want to put
molec.reset_charge()
molec.atoms[0].set_charge(-1)
molec

## Notice that we reset the charge, and added the -1 charge to the O atom di
## SPECIE and CELL charges/spins are always computed as the sum of atomic ch
## This means that, if you change an ATOM charge/spin, you're automatically
```

Out[53]:
```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = molecule
 Number of Atoms       = 3
 Formula               = H2-O
 Charge                = -1
 Spin (alpha - beta)   = 0
 Number of Parents     = 0
 Has Adjacency Matrix  = NO
 Has Bonds             = NO
```

In [54]:
```
## We reset the charge, since we're gonna use this molecule in Tutorial 7, a
## Basically, because it is not compatible
print(molec.check_spin_charge_compatibility())
## So we reset once again
molec.reset_charge()
```

```
False
```

In [55]:
```
## For SPIN, it works very similarly:
molec.set_spin_metals(2, debug=1)

## Logically, it won't find any metal atom in a water molecule, so it will c
## Most functions have a debug variable as input, with 1 'being verbose', ar
```

```
MOLECULE.SET_SPIN_METALS: there are no metals in this molecule
```

Loading [MathJax]/extensions/Safe.js

```
tely, we still have one of ABITEM's molecules in memory ("mol"). H
```

```
mol.set_spin_metals(2, debug=1)  ## 'mol' is one of ABITEM's molecules
```

```
MOLECULE.SET_SPIN_METALS: Preparing Spin Configuration for Specie H18-C20-N6
-S2-Fe
MOLECULE.SET_SPIN_METALS: Received spins=[2]
MOLECULE.SET_SPIN_METALS: Setting spin=2 to metal atom Fe
```

Out[56]: 2

Initial State (States are explained in Tutorial 3)

In [57]:
```
## Now we Create a State with the initial information, from which we will dr
## The name "initial" is not mandatory.
ini_state = molec.add_state("initial")    ## Here, the state is created, add
## As we mention in Tutorial 3, "initial" states (whatever the name) are not

## SPECIE sources only require geometry:
ini_state.set_geometry(labels, coords)

## CELL sources would also require the cell_vectors and/or cell_parameters,
#ini_state.set_cell()
```

Add Molecule as Source, and Save the system

In [58]:
```
## And source it to our system, giving it the name "geom1"
_ = test_sys.add_source('geom1',molec)
## Notice that the molecule has been added in the list of sources:
print(test_sys)
```

```
---------------------------------
   >>> SCOPE System Object >>>
---------------------------------
 Version                = 1.0
 Type                   = system
 Subtype                = system
 Source Path            = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Sources/water/
 System File Path       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Systems/water/
 System File Name       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Systems/water/water.npy
 Computations Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/7-Tutorial_7/Computations/water/

 Num Sources            = 1
     idx: type, name, formula
     0: specie, geom1, H2-O
```

In [59]:
```
## Trying to add new molecules with the same name won't work, to avoid dupli
_ = test_sys.add_source('geom1',molec, debug=1)
```

Loading [MathJax]/extensions/Safe.js

```
SYSTEM.FIND_SOURCE. Searching for source with geom1 in system with 1 source
s:
    geom1 specie
SYSTEM.ADD_SOURCE: Source with name 'geom1' already exists in system 'water'
SYSTEM.ADD_SOURCE: If you would like to Overwrite, specify overwrite=True
```

In [60]:
```
## Finally, save the system to a file. By default, it will use the path stor
test_sys.save()
```

Loading [MathJax]/extensions/Safe.js